University of Frontier Technology

Department of Educational Technology and Engineering (EdTE)

Faculty of Digital Transformation

Session: 2022-2023

Lab 10: Serverless Computing (AWS Lambda / Google Cloud Functions)

Submitted by:

Hafiza Hydar Hinu

2202043

Submitted to:

Sunjida Akter

Lecturer

Department of Educational Technology and Engineering (EdTE)

University of Frontier Technology, Bangladesh

Date:12July, 2026

Serverless Computing (AWS Lambda / Google Cloud Functions)

1. Objective

The main objectives of this experiment are:

1. To understand the basic concept of Serverless Computing.
2. To learn how AWS Lambda works without directly managing servers.
3. To create a simple serverless function using AWS Lambda.
4. To deploy and execute the function in the AWS environment.
5. To test the function and observe its execution output.

2. Theory

Serverless Computing is a cloud computing model where the cloud service provider manages the required servers, infrastructure, scaling, and maintenance. In this model, developers can concentrate on writing and deploying applications without having to manage the underlying servers manually.

Although it is called "serverless," physical servers are still used. However, their provisioning, maintenance, and scaling are handled by the cloud provider. Popular cloud providers that offer serverless services include AWS, Google Cloud, and Microsoft Azure.

The major features of Serverless Computing are:

- **No Server Management:** The cloud provider manages server provisioning, maintenance, updates, and scaling.
- **Event-Based Execution:** Functions are executed when a specific event occurs, such as an HTTP request, file upload, or database update.
- **Pay-as-You-Use:** Users are generally charged according to function execution and usage rather than paying for continuously running idle servers.
- **Automatic Scaling:** The platform automatically increases or decreases resources according to the workload.

AWS Lambda is a serverless computing service provided by Amazon Web Services. It allows developers to execute code in response to different events without creating or managing servers manually. Lambda functions can be developed using programming languages such as Python, Node.js, Java, and Ruby. These functions can be triggered through services such as API Gateway, S3 events, or test events from the AWS Console.

Similarly, **Google Cloud Functions** is another Function-as-a-Service (FaaS) platform that provides serverless function execution within Google Cloud.

3. Tools / Environment Used

The following tools and environment were used in this experiment:

- **AWS Management Console**
- **AWS Lambda Service**
- **Region:** Europe (Stockholm) – `eu-north-1`
- **Runtime:** Python
- **Source File:** `lambda_function.py`
- **Built-in AWS Code Editor / Amazon Q assistance**

4. Procedure

Step 1: Open AWS Lambda

First, log in to the AWS Management Console. Use the search bar to find the **Lambda** service and open it.

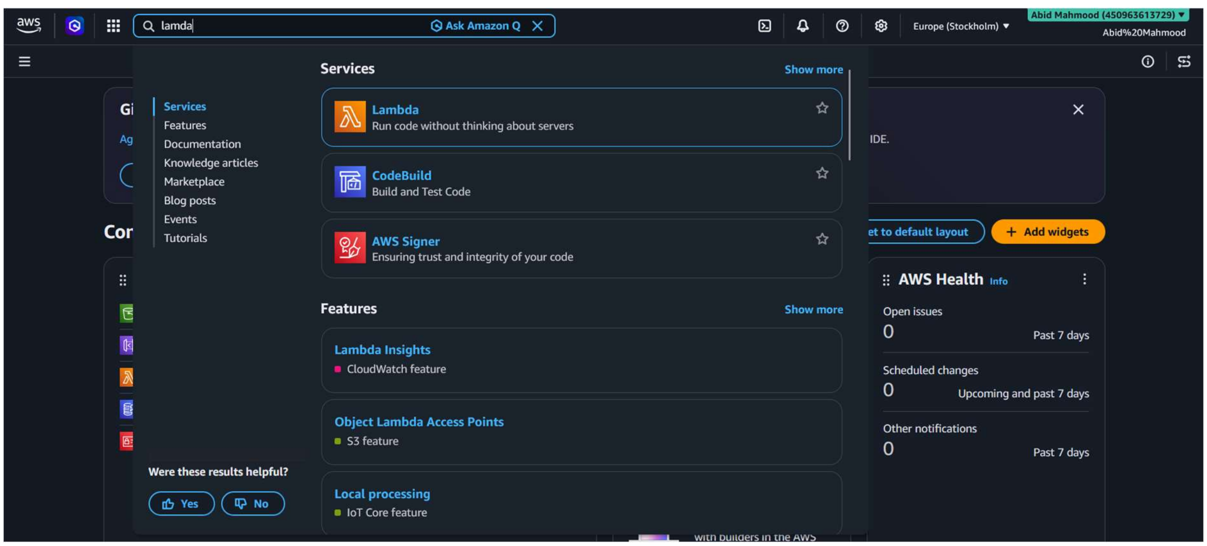

Fig. 1 – Searching for the Lambda service in the AWS Console.

Step 2: Create a Lambda Function

Click on **Create function**. Select **Author from scratch** and provide a suitable function name. In this experiment, the function name is **hello**. Then, select the required Python runtime and create the function.

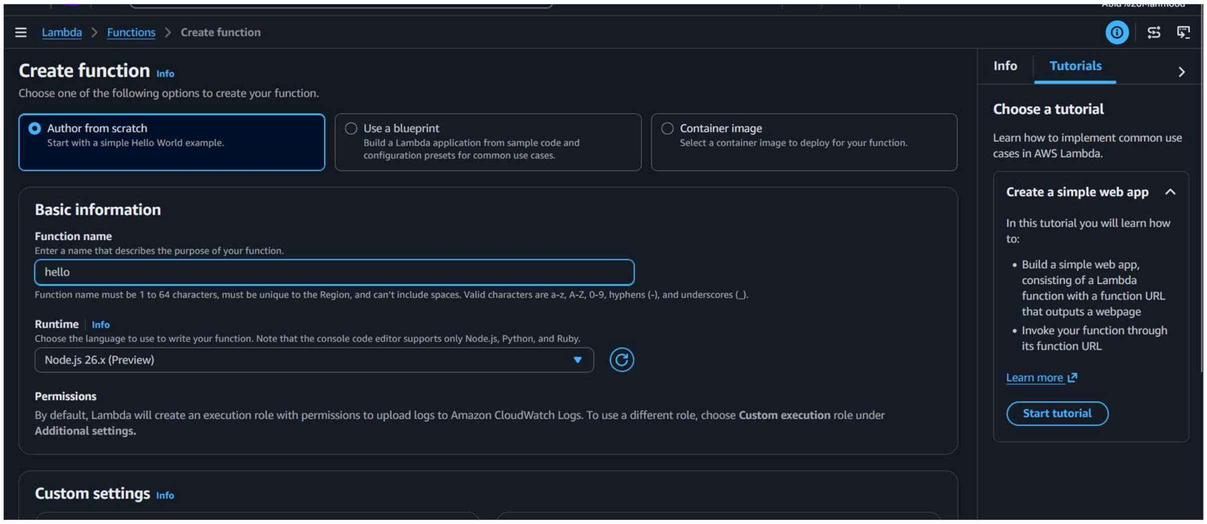

Fig. 2 – Creating a new Lambda function named "hello".

Step 3: Verify the Created Function

After the function is created successfully, it appears in the **Functions** list of AWS Lambda. This confirms that the Lambda function has been created and deployed successfully.

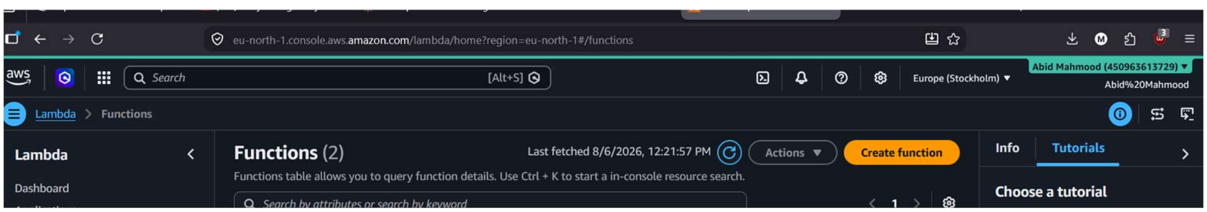

Fig. 3 – The "hello" function listed under Lambda Functions.

Step 4: Add and Test the Function Code

Open the `lambda_function.py` file from the built-in code editor. Write the required handler code and save the changes. Then, use the **Test** option to invoke the function with a test event.

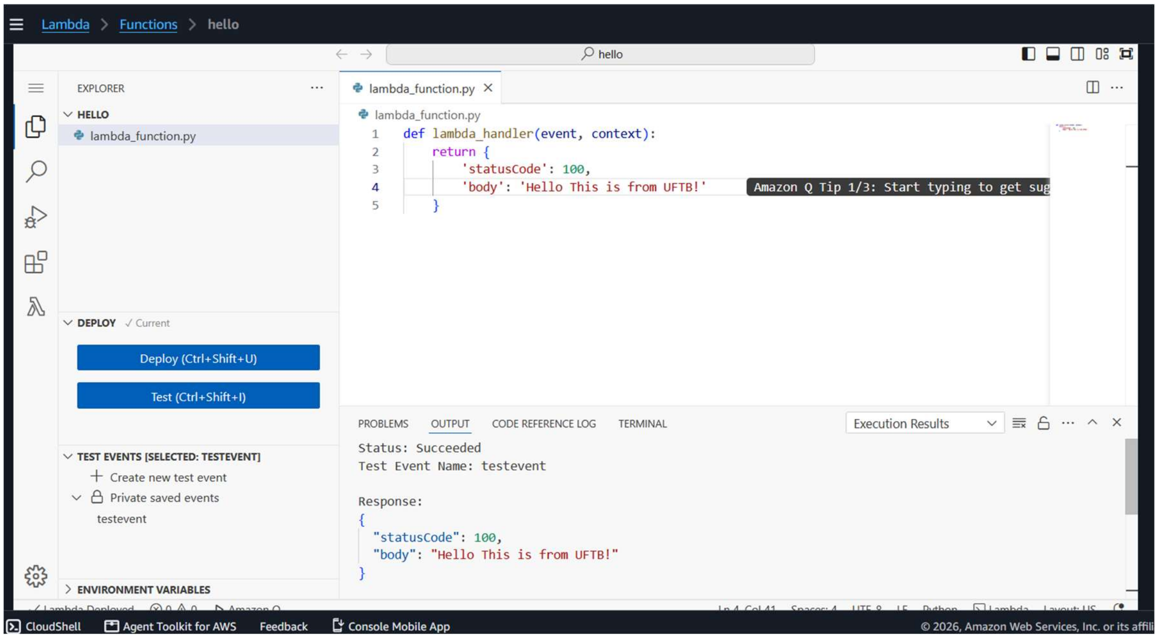

Fig. 4 – Lambda function code and successful test execution output.

5. Source Code

```
def lambda_handler(event, context):
    return {
        'statusCode': 100,
        'body': 'Hello This is from UFTB!'
    }
```

6. Result / Output

On invoking the function with the test event “testevent”, the execution succeeded and returned the following JSON response:

```
{
  "statusCode": 100,
  "body": "Hello This is from UFTB!"
}
```

This confirms that the serverless function was deployed and executed correctly on AWS Lambda without any manual server configuration.

7. Conclusion

In this experiment, a simple serverless function was successfully created, deployed, and tested using **AWS Lambda**. The experiment demonstrated how serverless computing allows developers to execute code without directly managing backend servers. The function was executed through a test event, and the AWS Console provided an easy environment for writing, deploying, and testing the code.

Therefore, this experiment helped demonstrate the basic working process and major benefits of serverless computing, including simplified infrastructure management, on-demand execution, and automatic scalability.